

Learning from Big Data: Tutorial 1.1

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Introduction

This tutorial focuses on applying Natural Language Processing (NLP) techniques for supervised learning. We begin by using the Naive Bayes Classifier (NBC) on review data to determine both the topic and sentiment of a review. Next, we apply VADER, a lexicon-based tool, to perform the same sentiment analysis task. Additionally, we will cover how to evaluate model performance using a confusion matrix, explaining how it visualizes predictions, can highlight incorrect classifications, and supports the calculation of key performance metrics used to evaluate our models.

1. Loading libraries

Before starting the tutorial, make sure you have all the required libraries properly installed. Simply run this chunk of code below.

```
# Required packages. P_load ensures these will be installed and loaded.
if (!require("pacman")) install.packages("pacman")
pacman::p_load(tm, nnet, dplyr, tidyr, ggplot2, reshape2, latex2exp, vader, caret, png, knitr)
```

2. Load the reviews and prepare the data

```
setwd("C:/Users/josef/Desktop/Learning-from-Big-Data-main/Learning-from-Big-Data-main/tutorials/input")
# Load the review data. Note that we are now using the fileEncoding parameter when
# calling read.csv() - this helps reading the review text correctly for further
# processing (by correctly interpreting the non-ASCII symbols)
# The ISO-8859-1 encoding is used to represent the first 256 unicode characters
reviews_raw <- read.csv('Reviews_tiny.csv', fileEncoding="ISO-8859-1")

reviews_raw <- reviews_raw %>%
  select(movie_name, review_code, reviewer, review_date, num_eval,
         prob_sentiment, words_in_lexicon_sentiment_and_review, ratio_helpful,
         raters,
         prob_storyline, prob_acting, prob_sound_visual,
         full_text, processed_text,
         release_date, first_week_box_office, MPAA, studio, num_theaters )
```

Texts labelled with content or sentiment are subsequently used to compute word likelihoods. The code chunk below loads training data from three content lexicons related to storyline, acting, and visual aspects. Sentences are assigned to topics like acting, storyline, or visuals for topic classification using NBC. Furthermore, content/sentiment likelihood data is loaded. These are from movie reviews labelled as positive or negative used in our sentiment analysis. Finally, the parameters capturing the priors for our NBC models are defined

```

# training data
dictionary_storyline <- read.csv2("storyline_33k.txt")
dictionary_acting    <- read.csv2("acting_33k.txt")
dictionary_visual    <- read.csv2("visual_33k.txt")

# TODO: Compute the word likelihoods from 3 content dictionaries (i.e., your training data).
# Here, I load a list of 100 words with fake content likelihoods and a list with
# 100 fake sentiment likelihoods. These are just examples. These 100-word lists
# are not to be used in your assignment. In your assignment, you are expected to
# compute the content likelihoods for all the words in the training data yourself.
likelihoods <- read.csv("example_100_fake_likelihood_topic.csv")

# TODO: Locate a list of sentiment words that fits your research question.
# This is available from the literature. For example, you may want to look at
# just positive and negative (hence two dimensions), or you may want to look at
# other sentiment dimensions, such as specific emotions (excitement, fear, etc.).
# The list of 100 words with fake likelihoods for sentiment used below is not to be used.
likelihoods_sentim <- read.csv2("example_100_fake_likelihood_sentiment.csv",header=TRUE,
                                sep=";", quote="\"",dec=".",fill=FALSE)

```

These lexicons are used as (a dictionary of) words that are associated with our content topics/sentiment

```

lexicon_content  <- as.character(likelihoods[,1])
lexicon_sentiment <- as.character(likelihoods_sentim$words)

# Setting our prior parameters
# We set out priors for the topics to 1/3 each because we have three topics
# (i.e. storyline, acting, and visual). Similarly, we set the priors for
# sentiment to 1/2 each because we have two sentiments (positive/negative)
prior_topic <- 1/3
prior_sent  <- 1/2

total_reviews <- nrow(reviews_raw)

```

3. Supervised Learning: Naive Bayes Classifier (NBC)

The Naive Bayes Classifier is a probabilistic model based on Bayes' Theorem used to predict the probability that a given input, in this case reviews, belongs to a particular category. Throughout this tutorial the categories we will be using are sentiment and topics. NBC begins with a prior probability for each class, which represents the initial belief about the likelihood of each category. Then, for every word in the input, the model calculates the likelihood of that word appearing given each class. Using Bayes' rule, it continuously updates the probability for each class as more words are considered. Finally, the class with the highest posterior probability is selected as the predicted category.

Below, the functions `Compute_posterior_sentiment` and `Compute_posterior_content` are displayed. These functions apply the Bayes rule to calculate posteriors.

Compute posterior sentiment function

This first function estimates the probability that the review expresses positive or negative sentiment.

```

Compute_posterior_sentiment <- function(prior, corpus_in , dict_words, p_w_given_c,TOT_DIMENSIONS ){

  output <- capture.output(word_matrix <-
                            inspect(DocumentTermMatrix(corpus_in,

```

```
control=list(stemming=FALSE,
             language = "english",
             dictionary=as.character(dict_words))
```

Check if there are any relevant words in the review, if there are, treat them. if not, use prior

```
if (sum(word_matrix) == 0) {

  posterior<-prior ; words_ <- c("")

} else{

  # Positions in word matrix that have words from this review
  word_matrix_indices <- which(word_matrix>0)
  textual_words_vec <- colnames(word_matrix)[word_matrix_indices]

  # Loop around words found in review
  WR <- length(word_matrix_indices) ;word_matrix_indices_index=1
  for (word_matrix_indices_index in 1: WR){

    word <- colnames(word_matrix)[word_matrix_indices[word_matrix_indices_index]]
    p_w_given_c_index <- which(as.character(p_w_given_c$words) == word)

    # Loop around occurrences / word
    occ_current_word <- 1
    for (occ_current_word in 1: word_matrix[1,word_matrix_indices[word_matrix_indices_index]])
    {
      # initialize variables
      posterior <- c(rep(0, TOT_DIMENSIONS))
      vec_likelihood <- as.numeric(c(p_w_given_c$pos_likelihood[p_w_given_c_index],
                                   p_w_given_c$neg_likelihood[p_w_given_c_index]))

      # positive - this is the first element in the vector
      numerat <- prior[1] * as.numeric(p_w_given_c$pos_likelihood[p_w_given_c_index])
      denomin <- prior %*% vec_likelihood
      posterior[1] <- numerat / denomin

      # negative - this is the second element in the vector
      numerat <- prior[2] * as.numeric(p_w_given_c$neg_likelihood[p_w_given_c_index])
      denomin <- prior %*% vec_likelihood
      posterior[2] <- numerat / denomin
    }
  }
}
```

The %*% sign above indicates matrix multiplication, which is beyond the scope of the course

For those interested: <https://www.mathsisfun.com/algebra/matrix-multiplying.html>

```
if (sum(posterior)>1.01) {
  ERROR <- TRUE
}

prior <- posterior

} # close loop around occurrences

} # close loop around words in this review
```

```

    words_ <- colnames(word_matrix)[word_matrix_indices]

  } # close if review has no sent words

  return(list(posterior_=posterior, words_=words_) )
}

```

Compute posterior content function

This second function determines the probability that the review pertains to each specific topic.

```

Compute_posterior_content <- function(prior, word_matrix, p_w_given_c , BIGRAM, TOT_DIMENSIONS){

```

Check if there are any relevant words in the review, if there are, treat them. If not, use prior

```

  if (sum(word_matrix) == 0) {

    posterior<-prior

  } else{

    # Positions in word matrix that have words from this review
    word_matrix_indices <- which(word_matrix>0)
    textual_words_vec    <- colnames(word_matrix)[word_matrix_indices]

    # Loop around words found in review
    WR <- length(word_matrix_indices) ;word_matrix_indices_index=1
    for (word_matrix_indices_index in 1: WR) {

      word <- colnames(word_matrix)[word_matrix_indices[word_matrix_indices_index]]
      p_w_given_c_index <- which(as.character(p_w_given_c$words) == word)

      # Loop around occurrences / word
      occ_current_word <- 1
      for (occ_current_word in 1:word_matrix[1,word_matrix_indices[word_matrix_indices_index]])
      {
        # initialize variables
        posterior <- c(rep(0, TOT_DIMENSIONS))
        vec_likelihood <- as.numeric(c(p_w_given_c$storyline[p_w_given_c_index],
                                       p_w_given_c$acting[p_w_given_c_index],
                                       p_w_given_c$visual[p_w_given_c_index]) )

        # storyline - this is the first element in the vector
        numerat <- prior[1] * as.numeric(p_w_given_c$storyline[p_w_given_c_index])
        denomin <- prior %*% vec_likelihood
        posterior[1] <- numerat / denomin

        # acting - this is the second element in the vector
        numerat <- prior[2] * as.numeric(p_w_given_c$acting[p_w_given_c_index])
        denomin <- prior %*% vec_likelihood
        posterior[2] <- numerat / denomin

        # visual - this is the third element in the vector
        numerat <- prior[3] * as.numeric(p_w_given_c$visual[p_w_given_c_index])

```

```

    denomin <- prior %*% vec_likelihood
    posterior[3] <- numerat / denomin

    if (sum(posterior)>1.01) {
      ERROR <- TRUE
    }

    prior <- posterior

  } # close loop around occurrences

} # close loop around words in this review

} # close if review has no sent words

return (posterior_= posterior )
}

```

NBC Sentiment Analysis Loop

Now that we have defined the functions for calculating posteriors, we can loop over the reviews and apply these functions to determine the sentiment and content posteriors for each review. Using the ‘Compute_posterior_sentiment’ function we defined, we can calculate the posteriors for sentiment for each review using NBC.

```

# Loop over each review
for (review_index in 1:total_reviews) {

  # Print progress every 100th review
  if (review_index %% 100 == 0) {
    cat("Computing content of review #", review_index, " \n", sep="")
  }

  # If the review is not empty, continue and calculate posterior
  if ( reviews_raw$full_text[review_index] != ""){

    # Assign the processed text of the non-empty review to text_review
    text_review <- as.character(reviews_raw$processed_text[review_index])

    # Reset the prior every iteration as each review is looked at separately
    prior_sent_reset <- c(prior_sent, 1 - prior_sent)

    # Pre-process the review to remove punctuation marks and numbers.
    # Note that we are not removing stopwords here (nor elsewhere - a point for improvement)
    corpus_review <- tm_map(tm_map(VCorpus(VectorSource(text_review)), removePunctuation),
                           removeNumbers)

    # Compute posterior probability the review is positive
    TOT_DIMENSIONS <- 2
    sent.results <- Compute_posterior_sentiment(prior = prior_sent_reset,
                                                corpus_in = corpus_review,
                                                dict_words = lexicon_sentiment,
                                                p_w_given_c = likelihoods_sentim,
                                                TOT_DIMENSIONS)
  }
}

```

```

words_sent      <- sent.results$words_
posterior_sent  <- sent.results$posterior_

reviews_raw$prob_sentiment[review_index] <- posterior_sent[1]
reviews_raw$words_in_lexicon_sentiment_and_review[review_index] <- paste(words_sent, collapse = " ")

}
}

```

NBC Content Analysis Loop

We also calculate the posteriors for the content each review using NBC.

```

# Loop over each review
for (review_index in 1: total_reviews) {

  # Print progress every 100th review
  if (review_index %% 100 == 0) {
    cat("Computing content of review #", review_index, " \n", sep="")
  }

  # If the review is not empty, continue and calculate posterior
  if ( reviews_raw$full_text[review_index] != ""){

    # Assign the processed text of the non-empty review to text_review
    text_review <- reviews_raw$processed_text[review_index]

    # Pre-process the review to remove numbers and punctuation marks.
    # Note that we are not removing stopwords here (nor elsewhere - a point for improvement)
    # put in corpus format and obtain word matrix
    corpus_review <- VCorpus(VectorSource(text_review))

    output <- capture.output(content_word_matrix <-
                              inspect(DocumentTermMatrix(corpus_review,
                                                            control = list(stemming=FALSE,
                                                            language = "english",
                                                            removePunctuation=TRUE,
                                                            removeNumbers=TRUE,
                                                            dictionary=as.character(lexicon))

    # Compute posterior probability the review is about each topic
    TOT_DIMENSIONS <- 3
    posterior <- Compute_posterior_content(prior= matrix(prior_topic, ncol=TOT_DIMENSIONS),
                                           content_word_matrix,
                                           p_w_given_c=likelihoods,
                                           TOT_DIMENSIONS)

    # Store the posteriors
    reviews_raw$prob_storyline[review_index] <- posterior[1]
    reviews_raw$prob_acting[review_index] <- posterior[2]
    reviews_raw$prob_sound_visual[review_index] <- posterior[3]

  }
}

```

```

}

Processed_reviews <- reviews_raw
View(Processed_reviews)

# Saves the updated file, now including the sentiment and content/topic posteriors.
# write.csv(Processed_reviews,"TestProcessed_reviews.csv" , row.names = FALSE )

```

4. Supervised Learning: VADER

VADER is a sentiment analysis tool that uses a lexicon specifically designed to evaluate the sentiment score of texts. Each word in the lexicon is assigned a sentiment score indicating whether it is positive, negative, or neutral, allowing us to score overall sentiments of texts based on the combined scores of individual words. Following a similar approach to the Naive Bayes Classifier, we calculate the sentiment for each review and add the results to our dataframe.

```

# Loop over each review
for (review_index in 1:total_reviews) {

  # Print progress every 100th review
  if (review_index %% 100 == 0) {
    cat("Computing VADER sentiment of review #", review_index, " \n", sep="")
  }

  # If the review is not empty, continue and apply VADER
  if (reviews_raw$full_text[review_index] != ""){

    # Assign the processed text of the non-empty review to text_review
    # Note that we have not removed punctuation, numbers, and stopwords (a point for improvement)
    text_review <- as.character(reviews_raw$processed_text[review_index])

    # Apply VADER
    vader <- get_vader(text_review)

    # store the VADER results in the dataframe
    reviews_raw$vader_pos[review_index] <- vader[["pos"]]
    reviews_raw$vader_neg[review_index] <- vader[["neg"]]
    reviews_raw$vader_neu[review_index] <- vader[["neu"]]
    reviews_raw$vader_compound[review_index] <- vader[["compound"]]

  }
}

Processed_reviews <- reviews_raw
View(Processed_reviews)

#write.csv(Processed_reviews,"VADER_Processed_reviews.csv" , row.names = FALSE )

```

5. Performance Measurement: Confusion matrix

A confusion matrix is a valuable tool for evaluating classification models. A confusion matrix helps us compare our model's predictions with the true values by summarizing the counts of correct and incorrect

predictions across different classes. In classification problems, we typically focus on the positive class-the one we're interested in predicting-and the negative class, which represents all other outcomes. Below, you'll find an example of a confusion matrix.

A confusion matrix consists of: 1. True Positives (TP): Correctly predicted positive cases. 2. True Negatives (TN): Correctly predicted negative cases. 3. False Positives (FP): Incorrectly predicted as positive (Type I error). 4. False Negatives (FN): Incorrectly predicted as negative (Type II error).

Using a confusion matrix we can calculate metrics to calculate the performance of our classification model. A commonly used metric is the specificity The formula for the specificity is given below. A more comprehensive overview of metrics can be found here: https://en.wikipedia.org/wiki/Confusion_matrix#Table_of_confusion

$$\text{Specificity} = \text{TN} / (\text{FP} + \text{TN})$$

Now imagine we have trained a model that predicts whether an incoming email into our inbox is spam or not. We could use such a model's predictions to evaluate its effectiveness. Below is an example of how such an evaluation might look like with artificial values for actual and predicted outcomes. So, these values are not based off a real model, but are arbitrarily chosen to showcase how to use a confusion matrix in R and calculate specificity.

Artificial actual and predicted values for a classification problem with classes 'spam' and 'not spam'

```
actual <- factor(c("spam", "not spam", "spam", "spam", "not spam", "not spam", "spam", "not spam", "not spam", "not spam"),
                 levels = c("spam", "not spam"))
predicted <- factor(c("spam", "not spam", "not spam", "spam", "not spam", "not spam", "spam", "spam", "spam", "not spam"),
                   levels = c("spam", "not spam"))

# We can display the confusion matrix
conf_matrix <- confusionMatrix(predicted, actual)
conf_matrix$table

##           Reference
## Prediction spam not spam
##   spam       3       1
##   not spam   2       4

# Calculating the specificity
specificity(predicted, actual)

## [1] 0.8
```

You might have noticed that the supervised models in the previous sections do not output class labels (such as 'spam'/'not spam'), but rather a probability. At this stage, we distinguish between soft predictions and hard predictions. Soft predictions are the probabilities that an observation belongs to the positive class, while hard predictions are the final class labels assigned to the observations. To convert soft predictions into hard predictions, we use a decision rule, such as applying a threshold or selecting the class with the highest probability.

6. Brain teaser: hard vs. soft predictions

A classifier gives each observation a probability for each of three content categories. Two ways to use it: keep the probabilities (soft), or round them to the winner (hard). Hard predictions are particularly useful when a ready-to-use prediction is required. However, making the hard prediction inherently loses the more detailed information on the predicted probabilities. For example, imagine that you are building an information databank for yourself at work, and would like to aggregate all the incoming emails that are useful to you. Here, the emails are first categorized as either useful or not useful. Then, if the email is judged as useful, the information contained in the email is passed to some storage unit. The problem becomes clear - here, you

literally lose information if your classifier misjudges an email. Thus, it starts to make more sense to perhaps keep the probabilities on usefulness, so that you yourself can judge the email's content if the usefulness of its content is uncertain.

To showcase this problem, we construct the following scenario. We imagine 3 categories and a classifier that can make errors. We ask you to compare the information that results from using either hard or soft predictions.

6.1 Synthetic data: three category likelihoods + ground truth (given)

Draw the likelihoods at random (a Dirichlet draw: three positive numbers normalised to sum to 1), then draw the true category from those very likelihoods. That is what a well-specified classifier means - when it says 0.7, the truth is that category 70% of the time.

```
set.seed(20250902)

n      <- 900                # number of observations
alpha <- c(1.0, 0.7, 0.5)    # category 1 is the frequent one, category 3 the rare one

# empty containers that we fill in observation by observation
p      <- matrix(0, nrow = n, ncol = 3)  # the three likelihoods of each observation
truth <- rep(0, n)           # the true category of each observation

for (i in 1:n) {

  # draw three positive numbers and rescale them so that they sum to one
  draws <- c(rgamma(1, alpha[1]), rgamma(1, alpha[2]), rgamma(1, alpha[3]))
  probs <- draws / sum(draws)

  # store them, and draw the true category using those same probabilities
  p[i, ] <- probs
  truth[i] <- sample(c(1, 2, 3), size = 1, prob = probs)
}

# the data we work with: three likelihood columns and the ground truth
dat <- data.frame(p1 = p[, 1], p2 = p[, 2], p3 = p[, 3], truth = truth)
head(dat)

# average likelihood per category, and the share of each category in the truth
round(colMeans(p), 3)
round(table(truth) / n, 3)
```

6.2 Task 1 - Hard predictions: argmax

Predict each observation's category as the column with the highest likelihood. Report the confusion matrix and the accuracy, and compare the accuracy with the majority-class baseline and with the average winning probability. Are the errors spread evenly over the three categories? If not, which way do they lean, and why?

```
# TODO: loop over the observations and store, for each one, the category with the
#       highest likelihood (which.max) and the value of that winning likelihood (max)

# TODO: confusion matrix of truth against your prediction: table(truth, pred)

# TODO: accuracy = share of observations you got right; compare it with the
#       majority baseline max(table(truth) / n) and with the mean winning probability
```

6.3 Task 2 - Soft predictions: keep the probabilities

When evaluating forecasts of probabilities, data scientists often make use of the Brier score. In our case, we ask you to score the probabilities constructed above with the multiclass Brier score, $(1/n) * \sum_i \sum_k (p_{ik} - y_{ik})^2$, where y is 1 if the prediction is correct. This is especially appropriate when evaluating our soft predictions, which unlike the hard predictions, take on continuous values.

Compare three forecasts: the probabilities as they are, the same model rounded to a 0/1 call on the argmax, and a baseline that reports the average likelihoods for every observation. Which wins, and what exactly is the rounded version being punished for?

Then use the output as a measure rather than a label: compare the true share of observations in category 1 with the mean of `p1` and with the share of hard predictions equal to 1. Which of the two estimators is unbiased, and can you explain the direction of the other one's error?

```
# TODO: build the hard forecast as a matrix of probabilities: a 1 in the predicted
#       column of each row, 0 elsewhere

# TODO: build the baseline forecast: colMeans(p) repeated for every observation

# TODO: write a function brier(forecast) that loops over the observations, writes the
#       truth of observation i as a 0/1 vector y, adds up (forecast[i, ] - y)^2, and
#       returns the mean over all observations

# TODO: the Brier score of the three forecasts, side by side

# TODO: compare mean(truth == 1), mean(p[, 1]) and mean(pred == 1)
```

6.4 Wrap-up

In two or three sentences: which reading of the output would you report as “model performance”, and which would you feed into a regression? Why is a biased aggregate a worse problem than a noisy one?